

Decoupling stock between two production stages

Part 2 · Categories of stock — solution report with data, method, results and charts

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1. Part 2 · Categories of stock — Decoupling stock between two production stages

1. Decoupling stock between two production stages

Two stages, each with mean capacity **100/day** and CV **25%**. Throughput is limited by the slower stage every day unless a buffer decouples them.

85.9 /day

Throughput, no buffer

99.7 /day

Throughput, buffer 400

16.2 %

Gain

Decoupling stock lets stages run independently

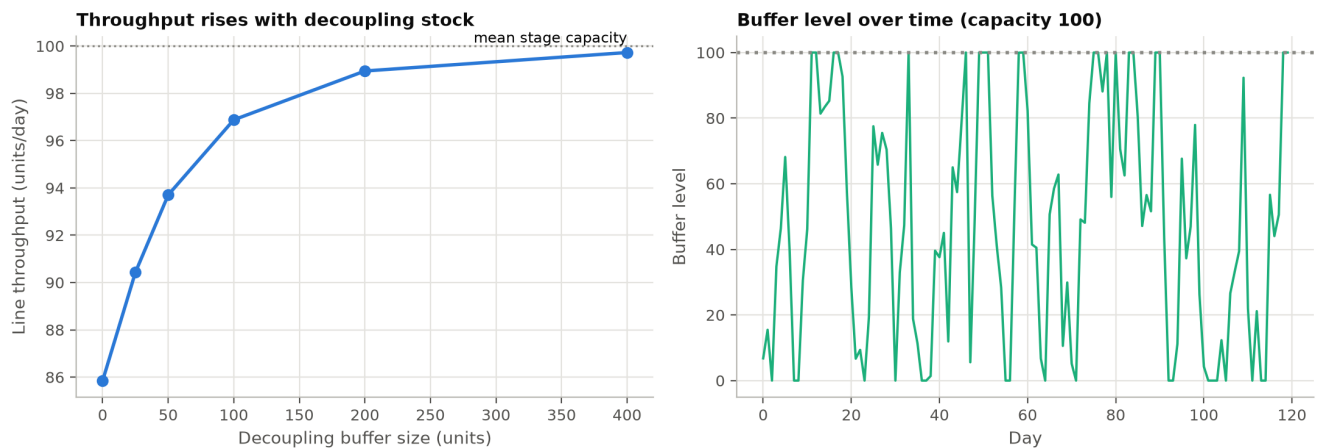


Figure 1.1 — Two-stage line simulation.

Throughput vs buffer size.

buffer capacity	throughput per day	utilisation stage2 pct	avg buffer stock
0	85.9	86.1	0.0
25	90.4	90.7	13.0
50	93.7	94.0	26.5
100	96.9	97.1	54.3
200	98.9	99.2	125.1
400	99.7	100.0	266.9